Summer 2026 Syllabus

Course Information

Course Number & Title: ENGR 13100 – Transforming Ideas to Innovation I

Credit Hours: 2.0

Modality: Face-to-Face & Synchronous-Online (see specific section)

Prerequisites: None

Sec #	CRN	Meeting Times	Classroom	Brightspace Page
001	25678	MWF 8:30AM – 10:20AM TTh 8:30AM – 11:20AM	LMBS 3285	Summer 2026 13100-001 SD
003	29814	MWF 8:30AM – 10:20AM TTh 8:30AM – 11:20AM	Synchronous Online	Summer 2026 13100-003 SD

Instructor Contact Information

Instructor contact information is provided in Brightspace along with graduate teaching assistant (GTA) contact information.

Course Description

A partnership between Schools and Programs within the College of Engineering, this course introduces students to the engineering professions using multidisciplinary, societally relevant content. Students will be developing engineering approaches to systems, generating and exploring creative ideas, and using quantitative methods to support design decisions. Explicit model-development activities (engineering eliciting activities, EEAs) engage students in innovative thinking across the engineering disciplines at Purdue. Students will experience the process of design and analysis in engineering including how to work effectively in teams. Students will develop skills in project management, engineering fundamentals, oral and graphical communication, logical thinking, and modern engineering tools.

Learning Outcomes

In ENGR 131, you will learn to:

- Develop and evaluate mathematical models to describe or predict outcomes associated with engineering decisions.
- Use a systematic design process to develop innovative solutions to engineering challenges, compare design alternatives, and gather and use evidence for decision making.
- Examine and analyze information from various engineering schools at Purdue to make informed decisions about your choice of engineering major.
- Practice making evidence-based engineering decisions.
- Work in diverse teams.
- Develop professional habits including professional communication, teamwork, and information literacy that will benefit you both as a student at Purdue and as a practicing engineer.

See the detailed learning objectives for this course in Brightspace.

Learning Resources, Technology, & Texts

There is no required textbook. You will use the following instructional technologies in ENGR 131:

- **Brightspace**: Within Brightspace, you will have access to course announcements, schedules, assignments, quizzes, exams, grades, feedback, and course resources. (Link to Brightspace).
- **CATME**: You will use CATME to submit information used for Team Formation and Peer & Team Evaluations (Link to CATME).
- **Gradescope**: You will submit homework assignments and exams to Gradescope for grading. You will access this through your ENGR 131 Brightspace section during the semester.
- **Messaging Tools**: Your section may use a messaging tool, such as Microsoft Teams, Slack or Discord. See your section's Brightspace, class slides, or email from your instructor/GTA regarding this.
- **MS Office**: Word, Excel, and PowerPoint. ENGR 131 requires Excel 2016 or newer. If you need an updated version, Office 365 is free to Purdue students. For more information, visit Purdue Office 365
- **Preferred browser**: Information Technology at Purdue (ITaP) recommends Google Chrome or Mozilla Firefox when accessing Brightspace. If you are using another browser or a mobile device, you may be unable to access some Brightspace content.
- **Purdue Email**: The teaching team will communicate using your Purdue email. Check your Purdue email regularly and often. Always respond or start an email to your instructor and/or GTA using your Purdue email.
- **Zoom**: Some meetings with an instructional team member may be handled via Zoom. Download the Purdue Zoom app on your laptop or PC. (Link to Purdue Zoom)

Your instructor may use additional technologies and software to facilitate interaction.

Course Structure

Each section of ENGR 131 is taught by a teaching team that includes an instructor, a graduate teaching assistant (GTA), undergraduate teaching assistants called Peer Teachers, and undergraduate graders. Course meetings are 110 minutes three days a week and 170 minutes two days a week, for 5 weeks. One section meets face-to-face, and the other section meets online during the same class time. Note that lecture recordings may not be available in your section. Other methods of delivery may be used as appropriate and will be communicated. You are required to attend during the scheduled class time (see also Attendance Policy).

Course Schedule

The course schedule will be provided to students on the Brightspace page. Schedule and assignments are subject to change throughout the 5-week course. Any changes will be updated in Brightspace.

Teaming

This is a project and team-based course. You will work in a collaborative learning environment on multiple assignments and a team project, involving authentic engineering situations. You will learn how to effectively work on a diverse team. Once team assignments begin, you are expected to have regular team meetings and work collaboratively on team projects and assignments. Your performance as a team member is part of your course grade. Failure to adequately and professionally engage in team activities may result in individual loss of credit on work that was turned in as a team (see Professional Expectations). Past ENGR 131 students have found their teaming experience in this course to be worthwhile and rewarding.

Assignments

ENGR 131 follows a flipped classroom model. Before class, you will watch videos and complete quizzes related to course topics. In class, you will work on activities, individually or in teams, related to the day's topics. Assignments and activities that are traditionally considered as homework are started in class. The teaching team will be

available during class time, so you can obtain help with understanding course topics and assignments from the various members of your teaching team.

In this course, the teaching team will assess your performance using the work you submit in four categories – pre-class quizzes, individual assignments, team assignments, and exams. Each of these assessments will include detailed instructions on what work to do, how it will be evaluated, when it is due, and how to submit it. It is your responsibility to follow the instructions on individual assignments, complete the work on your own, and submit it on time. Similarly, each team member is responsible for contributing a fair share of the work on a team assignment and making sure it is submitted on time.

Late Work Policy

On-time submission of your work is vital to successful performance in this course. Late work is only accepted as specified in the following cases:

- **Submitting late individual work without penalty**: You may submit late individual assignments without penalty only when you have an Office of the Dean of Students (ODOS) approved absence (see Exempt Absences). In these situations, arrange submission of your late work with your graduate teaching assistant (GTA) or instructor.
- **Submitting late individual work with a 25% penalty**: When late submissions are permitted, you may submit your work up to 24 hours late, but you will incur a deduction of 25% of the maximum possible points.
- **Team Assignments**: Any late submission of teamwork may be subject to a root cause analysis protocol, completed by your team, as facilitated by your GTA or instructor.
- **CATME Assignments**: Late submissions of the Team Formation Survey or Peer & Team Evaluations are not permitted.

No work may be submitted after the 24-hour late period without specific exception from your instructor.

Help Sessions

ENGR 131 provides help sessions as an opportunity for you to receive guidance in determining answers to specific questions on assignments or preparation for exams. See Brightspace for scheduled times and locations as well as general help session guidelines. Additional help is also available from your instructor; contact them for more information.

Grading Scale

Grades are assigned according to the percentage of available points earned as detailed in the table below. Note that to successfully pass ENGR 131, and move onto ENGR 132, your final grade must be a C- or better. There is no rounding of grades in ENGR 131. There are four categories of assignments that make up the final grade: Pre-Class Activities (9%), Individual Assignments (26%), Team Assignments (27%), & Exams (38%).

Grade	Grade Value
A	Greater than or equal to 94.0%
A-	Greater than or equal to 90.0% and less than 94.0%
B+	Greater than or equal to 87.0% and less than 90.0%
B	Greater than or equal to 84.0% and less than 87.0%
B-	Greater than or equal to 80.0% and less than 84.0%
C+	Greater than or equal to 77.0% and less than 80.0%

C	Greater than or equal to 74.0% and less than 77.0%
C-	Greater than or equal to 70.0% and less than 74.0%
D	Greater than or equal to 60.0% and less than 70.0%
F	Less than 60.0%

Grading & Feedback

The purpose of grading is to assess your understanding and utilization of the concepts taught in the course, and to provide you with feedback about the strengths and weaknesses evident in your work. Full credit may be awarded on items that are mostly correct even if the work still contains errors in understanding. Therefore, it is important that you not only check your score on a particular assignment or exam, but also review the feedback provided by the graders. This feedback will help you improve your understanding of the concepts being assessed and, in turn, improve your performance on future work.

Regrade Requests

Most grading for this course will happen in Gradescope. If you have concerns about how an assignment or exam problem was graded, use Gradescope's Regrade Request tool to submit a detailed description of the concern.

- Regrade requests are only open for a limited time after grades have been posted – your GTA will advise the class of each regrade window after assessments are graded.
- Your regrade request must be concise and professional. Please see Communication with the Teaching Team (under Professional Expectations below) for proper communication etiquette.
- If the regrade request is approved by the teaching team, your assignment or exam problem will be completely regraded. The updated grade may increase, decrease, or stay the same, depending on the outcome of the regrade process.

Professional Expectations

Each Professional Expectation (PE) in ENGR 131 reinforces the idea that everyone in our learning environment helps shape the environment so that it is positive and productive for all. This includes arriving to class on time and being prepared, listening and taking notes during the lecture, focusing on course activities during class, controlling your behavior to minimize distractions to those around you, and engaging with others in a respectful and professional manner.

Professional Expectations Deductions

The instructor of record may deduct points from your semester total for behavior that is unprofessional, disrespectful, or disruptive. This includes behavior directed toward classmates, instructors, GTAs, UGTAs, or any other course staff, as well as behavior that disrupts class activities or negatively impacts your team's ability to function and perform.

Communicating with the Teaching Team

When communicating with members of your ENGR 131 teaching team, your email must originate from your Purdue email account and include:

- your name
- ENGR 131, section number, and team number (once teams are assigned)
- topic (e.g., assignment name)
- a detailed description of your concern

For professional communication, make sure your email is:

- appropriately addressed to the recipient (e.g., not "Hey," but "Dear Professor"),
- includes a helpful subject line with ENGR 131 included (e.g., "ENGR 131: Question about A04"),
- written in complete sentences,
- specific (e.g., not "I have a question on the assignment" but "I have a question on part 2 of problem 3 in assignment A04"),
- concluded with an expression of appreciation for the reader's time or help.

Allow at least 24 hours for emails to be answered.

Attendance Policy

Maintaining contact with your instructor, class, and team is an important part of your success in the classroom. You are expected to attend classes and participate in the in-class activities. You and your team will both benefit from your participation.

You are responsible for:

- **Preparation**: be prepared for each class according to your instructor's directions. This includes completing the assigned pre-class reading/videos and pre-class quizzes (found on Brightspace), as well as other tasks as assigned.
- **Punctuality**: arrive on time for class and be prepared to participate.
- **Participation**: due to the team-oriented nature of this course, your consistent and enthusiastic participation in all parts of the course is important. Failure to participate in class will be considered equivalent to an unexcused absence.

Absence-Related Professional Expectations

- If you know you will miss a class, you must notify your instructor and GTA as soon as possible. Advance notice is expected except in emergencies.
- If you need an extension on an assignment due to an absence, a 24-hour extension is typical. However, unless the absence is excused by ODOS, extensions are **not automatic** and must **be requested in writing** before the assignment deadline.
- Communication with your teammates is a professional courtesy. This helps you stay informed about class topics and coordinate teamwork.
- You are responsible for catching up on all information presented in class in preparation for the next class.

Exempt Absences

- **Absences excused by the Office of the Dean of Students (ODOS)**. Under academic regulations, excused absences may be granted by ODOS for cases of grief/bereavement, military service, jury duty, parenting leave, or emergent medical care.
- **Chronic/long-term illness,** or other emergencies with documentation. In rare cases, instructors may designate absences as exempt if there is valid documentation, but the situation is not covered by ODOS or by Disability Resource Center (DRC) accommodations.
- **Other situations** may be excused at the **discretion of your instructor**, such as University-sanctioned athletics, bands, & orchestras.

Non-Exempt Absences

- Each student is allowed up to **three non-exempt absences** during regular class periods **without penalty**. These absences may occur for any reason (e.g., minor illness, travel, club/event participation) and may occur during any class period. See *Exam Period Absence* for make-up exam policies and restrictions.

Exam Period Absence

- Students with an **exempt absence** during an exam period may complete a make-up exam.
- Students with a **non-exempt absence** during an exam period may complete a make-up exam **only if**:
    - The absence was communicated in advance, and
    - The absence was either due to an immediate need (e.g., illness) or was pre-approved in writing by the instructor (e.g., travel or activity participation).
- Students who miss an exam without any prior communication will receive a **zero** on the exam.
- Make-up exams must be scheduled **outside regular class time**. It is the student's responsibility to coordinate with the instructor to schedule the make-up exam **within one week** of the original exam date.

Attendance Record Procedures

The teaching team will take attendance shortly after class begins. You will be marked **present** if you are seated with your team at the start of class. If you arrive late, you must quietly check in with the teaching team to ensure your attendance is recorded correctly.

You are responsible for regularly checking your attendance record and reporting any discrepancies to the teaching team promptly.

Category	Definition	Attendance Record	Non-Exempt Absence Count
Exempt	Did not attend class, with ODOS excused absence (or similar exemption)	E	--
Present	Seated with team at the start of class.	P	0
Tardy	Late arrival (10+ mins after class start) or early dismissal (10+ mins before class end) not approved by the teaching team.	T	0.5
Absent	Did not attend class.	A	1

Your **non-exempt absence total** is the sum of your tardy and absence counts. You are allowed a non-exempt absence count **up to three** without penalty. Each additional non-exempt absence or tardy will result in the following grade penalties:

- Each additional non-exempt absence (A) = -20
- Each additional tardy (T) = -10

If your **non-exempt absence total exceeds six**, you will **fail the course**, regardless of your grade.

Examples:

Example 1: A student has 3 non-exempt tardies and 2 non-exempt absences.

- Non-exempt absence count = 3.5; 10-point grade deduction.

Example 2: A student has 0 non-exempt tardies and 5 non-exempt absences.

- Non-exempt absence count = 5; 40-point grade deduction.

Example 3 A student has 3 exempt absences and 1 non-exempt absence.

- Non-exempt absence count = 1; No grade deduction.

Academic Integrity

You are a member of the Purdue community – a community that values integrity. Academic integrity is your pledge, as a member of the Purdue Engineering community, to uphold high standards of responsibility and honor in all your academic work. Your integrity is just as crucial to your engineering career as your technical knowledge. Academic dishonesty harms both. Build a strong foundation for ethical work and honorable standards as an engineer in your undergraduate years.

Your work in this class should fulfill Purdue's student-created honor pledge: *"As a Boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together – We are Purdue."*

To maintain academic integrity, comply with course and University policies. You will be held to the standards explained in these documents and to university regulations for student conduct.

- Statement of Integrity and Code of Conduct (link)
- Academic Integrity and You: Undergraduate Edition (link)

You will be assessed in this course using individual and team submissions. When you work on individual assignments, your submissions must represent your unique work. For team assignments, all submissions must originate from the team, reflecting the team's collective effort.

Submitted work will be periodically checked for various forms of academic dishonesty.

You are expected to submit solutions that are your own creation. You, individually or in a team, must design and develop your solutions without copying from or sharing with any outside sources (including, but not limited to, peers, electronic sharing sites, or generative AI tools). Failure to submit your own unique solutions undermines the learning process and violates the trust and expectations set by Purdue and the engineering community.

Good Collaboration and Contributor Acknowledgment

You are encouraged to discuss course concepts and assignments with others in a way that enhances your learning. However, all work you submit must represent your own effort (for individual assignments) or your team's effort (for team assignments). If you receive substantial help from someone that directly influences your approach, design, or solution, you must list them as a contributor.

You are expected to work independently on all individual assignments. Any work that appears to be substantially similar to another student's submission may be investigated as academic dishonesty, even if both students cite each other.

Use of AI

Generative AI is a tool with strengths and weaknesses, just like any other tool. Good engineers use tools to strengthen their problem-solving and critical-thinking skills. They understand their tools and use those tools ethically. Before you use generative AI, you need to understand how it can help you and what its limitations are.

General policy for generative AI use in this course: When you work on individual assignments, your submissions must represent your unique work. For team assignments, all submissions must originate from the team, reflecting the team's collective effort. Originality in work not only demonstrates a student's understanding and mastery of the material but also ensures a fair and equitable assessment of their skills and knowledge. It serves as a fundamental measure of a student's ability to apply what they have learned in the course to solve specific problems.

You are responsible for using generative AI in an ethical manner that supports and enhances your learning. The course academic integrity policy provides you with more detailed information about how to use generative AI appropriately. You must be familiar with and abide by these policies. They are summarized below.

You are PROHIBITED from using generative AI to undermine or devalue the learning experience for yourself or your classmates.

- Never ask generative AI to do a problem for you; always attempt your own solution before asking AI for help.
- Never plagiarize the output from a generative AI tool.
- Use of generative AI during an exam is strictly prohibited.
- Never submit any part of any course materials, which are copyrighted and cannot be shared outside the classroom without permission, to a generative AI platform.

Avoiding Self-Plagiarism

In this course, we uphold the highest standards of academic integrity, which includes avoiding self-plagiarism. Self-plagiarism occurs when a student reuses their own previously written work in a new context without proper citation. This can create a misleading impression of the student's understanding and effort.

To avoid self-plagiarism, please adhere to the following guidelines:

- Original Work: Ensure that all work submitted for this course is original and created specifically for each assignment. Reusing code or written work from previous assignments, semesters, or courses is not acceptable unless explicitly permitted by the instructor.
- Transparency: Be transparent with your instructor about your work. If you're unsure whether reusing a piece of your own work might constitute self-plagiarism, ask your instructor for guidance.

Remember, the goal of each assignment is to enhance your learning and understanding. Each piece of work you submit should reflect your current knowledge and skills. Avoiding self-plagiarism ensures that you are always pushing forward in your learning journey.

Material Copyrights

The ENGR 131 materials and their notes are copyrighted or derivatives of copyrighted materials and should not be sold, bartered, or posted on sites such as Course Hero, Chegg, and Quizlet in whole or in part without express permission from your instructor and the Associate Head of First-Year Engineering.

Usage of Recordings

Course recordings may be made available for students and should only be used for reviewing the content of the class. Course recordings may not be shared outside of ENGR 131 and shall not be uploaded to any external sites.

Academic Dishonesty

Academic integrity is one of the highest values that Purdue University holds. Incidents of academic misconduct in this course will be addressed by the course instructor. Any violation of course policies relating to academic integrity will result minimally in a failing or zero grade for that assignment/exam, and at the instructor's discretion may result in a failing grade for the course. Incidents of academic misconduct will be referred to the Office of Student Rights and Responsibilities (OSRR) for review, where university penalties may be considered.

Commitment and Support

The ENGR 131 instructional team is committed to supporting your success with integrity. If you are unsure whether a particular action violates the course policy, ask your instructor or graduate teaching assistant. If you witness or suspect academic dishonesty, report it promptly. When in doubt—ask. Integrity matters.

Academic Support

The Helen Bass Williams Academic Success Center, provides a variety of proactive, practical and approachable academic support services for you to strengthen your approaches and strategies for learning, including study skills consultations, peer coaching, workshops, and online handouts. Visit the ASC website for more information and to access resources.

University Policies

Nondiscrimination Statement

Purdue University is committed to maintaining a community that recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. A hyperlink to Purdue's full Nondiscrimination Policy Statement is included in our course Brightspace under University Policies.

Accessibility

Purdue University is committed to an inclusive and welcoming experience for all students. We strive to make learning experiences accessible to all participants. The Disability Resource Center (DRC) provides services, resources, and programs to facilitate equal access for disabled students, resulting in their full participation in curricular and co-curricular offerings.

If you anticipate or experience physical or academic barriers based on disability, you are encouraged to contact the Disability Resource Center (drc@purdue.edu or 765-494-1247). If you are eligible for academic accommodations because you have a documented disability that will affect your work in this class and/or for an exam, you will use your DRC Course Accessibility Letter and work with your instructor as soon as possible to discuss your needs.

Students with disabilities whose DRC Course Accessibility Letter includes test accommodations will work through your instructor and the FYE Instructional Support Team, so they can ensure proper accommodations are made. In some cases, exams may need to be scheduled with Purdue Testing Services (PTS) at https://www.purdue.edu/studentsuccess/testing-services/accommodated-testing/student.php. You must do this at least four calendar days before the exam date listed on the syllabus. The FYE Instructional Support Team will provide PTS with your exam information. PTS will administer and proctor the exam for you. Students must inform the Instructor and GTA immediately of cases where PTS does not have space so that the FYE Instructional Support Team can make other arrangements. Students who fail to follow this process and these deadlines risk not being able to have their accommodations for that exam. If we can do anything further to support your access to and engagement with materials, please let your instructor know.

Mental Health/Wellness Statement

- **If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students.** If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS office on the second floor of the Purdue University Student Health Center (PUSH) during business hours.
- **If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try** Therapy Assistance Online (TAO), a web and app-based mental health resource available courtesy of CAPS. TAO is available to you at any time by creating an account on the TAO Connect website, or downloading the app from the App Store or Google Play. It offers free, confidential well-being resources through a self-guided program informed by psychotherapy research and strategies that may aid in overcoming anxiety, depression and other concerns. It provides accessible and effective resources including short videos, brief exercises, and self-reflection tools.
- **If you need support and information about options and resources**, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 am- 5 pm.
- **If you find yourself struggling to find a healthy balance between academics, social life, stress**, etc., sign up for free one-on-one virtual or in-person sessions with a Purdue Wellness Coach at RecWell. Student

coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can be done on BoilerConnect.

Basic Needs Program

Any student who faces challenges securing their food or housing, transportation support, physical, mental, and behavioral health services, childcare resources, and technology access, and believes this may affect their performance in the course is urged to contact basicneeds@purdue.edu. Please visit the Basic Needs Program website for additional information.

Emergency Preparedness

Purdue University has an Integrated Emergency Management Plan (IEMP). This plan includes procedures, processes, and plans for responding to an emergency. Visit the Emergency Preparedness website for more information. Keep your cell phone on to receive a PurdueALERT text message. When on campus, it is also important to be familiar with the emergency response procedures for your location and follow the instructions from emergency response personnel (police, fire department, etc.) when they are present.

Procedures specific to the ENGR 131 classrooms are below:

- **General Emergency:** For ANY emergency, call 911 (fire, medical emergency, etc.).
- **Fire Alarm or Evacuation:** Exit the building using the stairs. When exiting the building, do not use the elevator. Once outside the building, stay clear of all emergency vehicles and personnel. Meet your classmates north of Lambertus Hall (LMBS) by Mechanical Engineering (ME).
- **Shelter in Place:** Could occur due to various situations such as a tornado, accidental release of toxic chemicals, an active threat (e.g., shots fired on campus).
    - **Tornado:** Proceed down the stairs to the basement of Lambertus Halls (LMBS) Be prepared to sit (or kneel) on the floor, face a wall, and cover your head.
    - **Other situations:** The course of action would depend on the situation. It is recommended that students remain in the classroom and wait for further instructions.

Course Adjustments

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructor or GTA via email or phone. You are expected to read your @purdue.edu email on a frequent basis.